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MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MID TERM EXAMINATION, JULY-DEC -2023

COURSE: M.TECH

SEMESTER: I

BRANCH: CN

SUBJECT: Wireless Networking

SUBJECT CODE: CN-514

DATE: 24/11/2023

MAX MARKS:20

EXAM TIME: 09:00AM to 10:30PM

NOTE: All questions are compulsory

Q.No.	Question	Marks	CO
1	a) Write a short note on AMTS. b) Write a short note on D-AMPS.	5	CO1
2	Hetnet and eICIC is used in which release of LTE.? Elaborate it with their work..	5	CO1
3	a). You are streaming a video on your wifi network and it requires a constant 5mbps of BW. If there are three other devices simultaneously using the network, each requiring 2mbps. Can your network handle this load if you have a 20mbps internet connection?	5	CO2
	b). What are some best practices for securing a public wifi network?		
4	a). What are the challenges to implement wireless communication over the diverse geographic distribution in rural areas? b). How does CA (carrier aggregation) helps mobile to fast internet speed?	5	CO2

Name of the Student.....SAURAV..... Scholar Number.....06.....

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

EXAMINATION: _____ MONTH and YEAR: Dec 2023

Course: M.Tech Semester: I Branch: CN Subject Code: CN 514

Subject Name: Wireless Networking Maximum Marks: 50

Duration: 03:00 hours Date: 20/12/23 Time: 09:30AM to 12:30 PM

Note: *Take suitable assumptions in numerical if required*

Q. No.	Questions	Marks	COs
Q.1	A) What are the different wireless communication standards? list out various WLAN standards with typical features of each?	3	CO1
	B) How cell sectoring can be used to increase the capacity of a cellular system? Explain what are the drawbacks of cell Sectoring.	3	CO1
	C) Explain various bluetooth connection establishment states. Explain with complete flow diagram	4	CO2
Q.2	A) Given a cellular system in which there are a total of 1001 duplex radio channels available for handling traffic. It is also given that the area of a cell is 6km^2 and the area of the entire system is 2100 km^2. calculate the system capacity if the cluster size is 7 and also find the number of times that the cluster has to be replicated over the entire service area of cellular system.	5	CO3
	B) Give the technical requirements of IEEE 802.15.3a physical layer with respect to data rates, power consumption,power management mode, scalability and interference etc.	5	CO3
Q.3	A) Discuss about TDMA(time division multiple access) technique used in wireless communication. if GSM uses a frame structure where ach frame consists of 8 time slots, and each time slot contains 156.25 bits (assuming a normal GSM time slot consists of 6 trailing bits, 8.25 guard bits, 26 training bits and two traffic busses of 58 bits of data), and the data is transmitted at 270.833kpbs in the channel, then find 1)the time duration os slot 2) the time duration of a frame 156 3) frame efficiency.	6 4	CO2
	B) Why is network management design a critical issue in wireless sensor networks? Give reasons		CO1

Q.4	A) Consider a cellular system with 416 radio channels available for handling traffic. suppose 21 of their channels are designed to control channels. let the average channel holding time of a call be 3 minutes, the blocking probability during busy hours be 2% and the frequency reuse factor be 9. 1) determine the number of calls per hour 2) Determine the signal to channel interference ratio in DB	6	CO4
	B) Discuss in detail how packets are routed in MANET using dynamic source routing algorithms	4	CO3
Q.5	A) Explain and compare the working mechanism of both destination sequenced distance vector (DSDV) and dynamic source routing protocols when applied on a mobile adhoc network scenario.	5	CO5
	B) differentiate between- 1) eICIC and FeICIC 2) MIMO and MIMO advanced	5	CO4

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Name of the Student: Rohan Dubey Scholar Number: 24121011118

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, BHOPAL
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
EXAMINATION: MID TERM **MONTH and YEAR: Nov 2024**

Course: M. Tech **Semester:** I **Branch:** CN
Subject Code: FCN-512 **Subject Name: Wireless Networks**
Maximum Marks: 40 marks
Duration: 60 minutes **Date:** 19-11-2024 **Time:** 09:30 – 10:30 am

Note: 1. Attempt all questions, **2.** Assume necessary data if required and mention it clearly.

Q. No.	Questions	Marks
1	What are the different technologies used in 1G/2G/3G/4G/5G and 6G? What are the approximate data rates and modulation techniques used in these technologies?	**(8)**
2	Consider a wireless communication system operating at a frequency of 2.4 GHz. The transmitter power P_t is 20 dBm, the gains of the transmitting and receiving antennas are G_t=10 dB and G_r=5 dB, respectively. The receiver power P_r is measured to be −50 dBm. Assuming free-space propagation, calculate: (i) path loss PL in dB and (ii) the distance d between the transmitter and receiver	**(8)**
3	The cellular system is having total available spectrum of 25 MHz and channel bandwidth per user is 25 kHz for 7-cell reuse pattern. Other parameters are: 1. Number of cells in the network: 210 2. Call duration per user: 120 seconds 3. Handoff dwell time: 2 seconds 4. Probability of call interference during a handoff: 0.02 (i) Find the total system capacity (ii) Determine the number of channels available per cell. (iii) Estimate the average interference duration per call if 5% calls experience interference. (iv) Determine the percentage of time spent on handoff during a call.	**(8)**
4.	What are channel assignment strategies? Explain AMPS Dupoly channel assignment strategy for minimizing adjacent channel interference.	**(8)**
5.	Write short notes of two:- (i) Traffic engineering (ii) Mobile radio propagation characteristics (iii) Hidden terminal and exposed terminal problem in WLAN (iv) Comparison of narrow spectrum and spread spectrum	**(8)**

**********END**********

Name of the Student..... Scholar Number 24121011118

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, BHOPAL
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
EXAMINATION: END TERM **MONTH and YEAR: Dec 2024**

Course: M. Tech **Semester:** I **Branch:** CN
Subject Code: FCN-512 **Subject Name:** Wireless Networks
Maximum Marks: 50 marks
Duration: 2 Hrs **Date:** 16-12-2024 **Time:** 09:30 – 11:30 am

Note: 1. Attempt all questions, **2.** Assume necessary data if required and mention it clearly.

Q. No.	Questions	Marks	
1	Describe the O-RAN architecture and the features of 5G wireless communication.	(5)	CO4
2	What is the technology used in MIMO? How does massive MIMO enhance the bandwidth and data rate in 5G communication?	(5)	CO4
3	Describe different media access techniques in wireless communication. How OFDM adapt or enhance media access techniques to meet modern requirements like ultra-low latency and high reliability in 5G/6G technologies?	(6)	CO3
4	What are the different media access control techniques used in WLAN? Describe CSMA/CA with Binary Exponential Backoff (BEB) algorithm.	(6)	CO3
5	What are the characteristics of normal and abnormal Hand off? A mobile user travels at 50 km/h through a cell with a radius of 1 km. The dwell time is defined as the time the user remains within the cell boundary. • Calculate the dwell time for the mobile user in the cell. • If a handoff requires 5 seconds to complete, determine if there is sufficient time for a seamless handoff.	(6)	CO2
6	Explain the concept of CDMA. Consider a CDMA system in which user A & B have the code cA= <1,-1,-1, 1, -1, 1> and cB= < 1, 1, -1, -1, 1, 1> respectively. (i) Show the output of the receiver if A transmits data bit 1 in fist case and & 0 next. (ii) Show the output of the receiver if A tries to read the B's data.	(6)	CO3
7.	Consider a cellular system in which there are total of 1001 radio channels available for handling traffic. Suppose the area of the cell is 6 km^2 and the area of the entire system is 2100 km^2 . a) Calculate the system capacity if the cluster size is 7.	(8)	CO1

	b) How many times would the cluster of size 4 have to be replicated in order to approximately cover the entire cellular area? c) Calculate the system capacity if the cluster size is 4. d) Does decreasing the cluster size increase the system capacity? Explain		
8	Write short notes any two: - (i) Mobile radio propagation models (ii) Cognitive Radio (iii) Methods to increase capacity in Cellular Networks (iv) Hidden and Exposed terminal in WLAN	8	CO2 & 3

**********END**********